A11 / ELECTRICAL DRAFT — NOT A MANUFACTURING RELEASE

All decoding/DSP stays on Raspberry Pi; processed output goes to amplifier by USB.

ARC candidate + four-input selector + LVDS. Unrouted; DD+ and firmware not qualified.

power-control

File: power-control.kicad_sch

spdif

File: spdif.kicad_sch

selection-lvds

File: selection-lvds.kicad_sch

speakers

File: speakers.kicad_sch

arc-receiver

File: arc-receiver.kicad_sch

arc-connector

File: arc-connector.kicad_sch

arc-power

File: arc-power.kicad_sch

arc-decoupling

File: arc-decoupling.kicad_sch

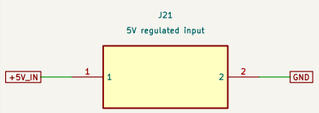
See requirements.md, sourcing.md and verification.md before using the model.

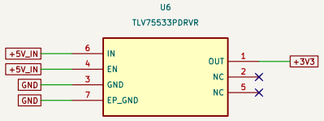
Sheet: /
File: rear-io-A11.kicad_sch

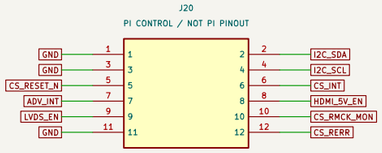
Title: Open Cinema rear I/O electrical study

Size: A3	Date: 2026-11-04	**Rev: A11 DRAFT**
KiCad E.D.A. 10.0.5		Id: 1/9

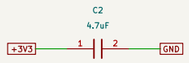
A11 ARC STUDY — 5V power and Pi control



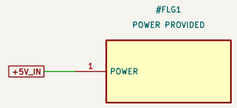


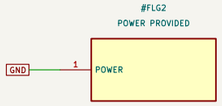






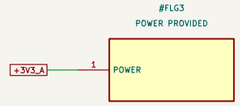
















5V regulated input only. U6 is now WSON-6 with grounded thermal pad.

R1/R2 DNP: use one I2C pull-up set at HAT; bus <=100 kHz. Both boards powered together.

J20 changed: pin7 ADV_INT, pin8 HDMI_5V_EN. Boot: CS reset low, LVDS and HDMI5V off.

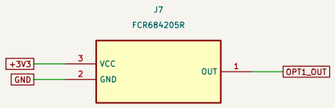
I2C: CS8416 0x20, ADV7511 0x39 (7-bit). Auxiliary ADV maps must not collide.

Sheet: /power-control/
File: power-control.kicad_sch

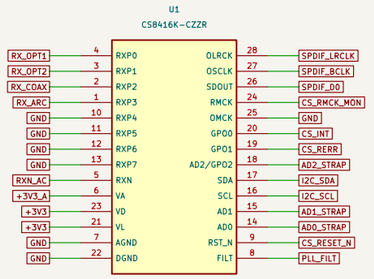
Title: Open Cinema — power-control

Size: A3	Date: 2026-11-04	**Rev: A11 DRAFT**
KiCad E.D.A. 10.0.5		Id: 2/9

A11 ARC STUDY — optical/coax/ARC shared selector







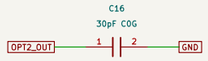






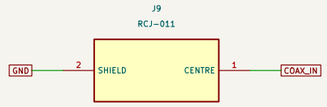






































RXP0 OPT1 / RXP1 OPT2 / RXP2 coax / RXP3 ARC. No resampling or codec decoding here.

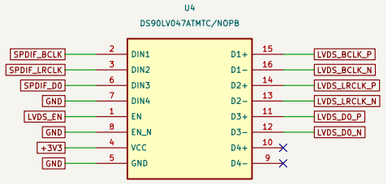
OMCK unused: carrier-rate measurement via RMCK needs Pi/HAT validation. Coax ESD remains open.

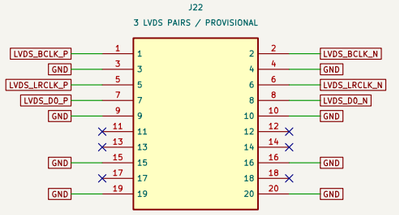
Sheet: /spdif/
File: spdif.kicad_sch

Title: Open Cinema — spdif

Size: A3	Date: 2026-11-04	**Rev: A11 DRAFT**
KiCad E.D.A. 10.0.5		Id: 3/9

A11 ARC STUDY — three active LVDS pairs







CS8416 owns source selection. A10 muxes, second transmitter and external eARC header removed.

LVDS carries BCLK / LRCLK / DATA0. J22 retains old pair positions; unused lanes are NC.

Pi is clock consumer. Receiver-side 100R termination and capture timing still need qualification.

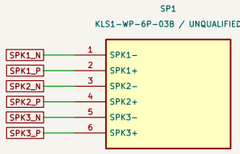
Disable LVDS and stop capture before changing source. Never play encoded words as PCM.

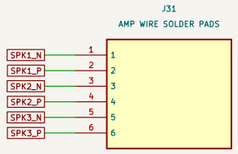
Sheet: /selection-lvds/
File: selection-lvds.kicad_sch

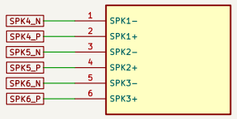
Title: Open Cinema — selection-lvds

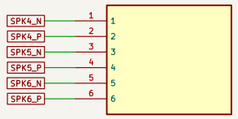
Size: A3	Date: 2026-11-04	**Rev: A11 DRAFT**
KiCad E.D.A. 10.0.5		Id: 4/9

A11 ARC STUDY — six independent speaker pairs









Six independent BTL pairs: speaker negatives must never share a return or connect to ground.

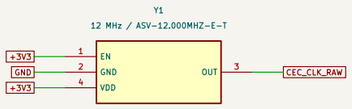
KLS clip current rating and exact supply remain unqualified. No fabrication/order release.

Sheet: /speakers/
File: speakers.kicad_sch

Title: Open Cinema — speakers

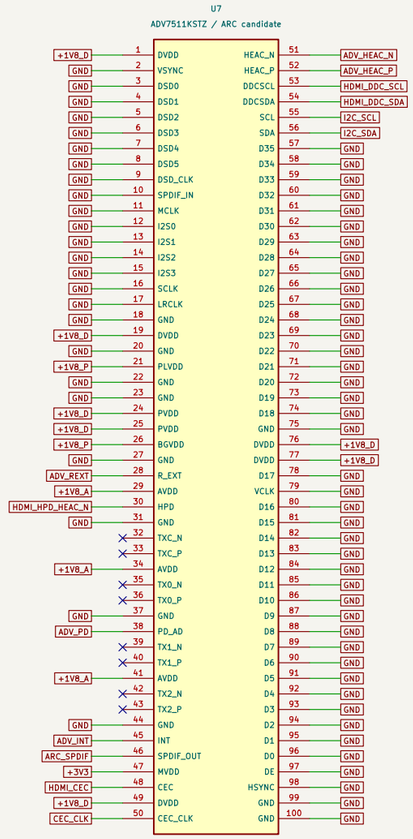
Size: A3	Date: 2026-11-04	**Rev: A11 DRAFT**
KiCad E.D.A. 10.0.5		Id: 5/9

A11 ARC STUDY — ADV7511KSTZ (100-pin LQFP)















All 100 pins represented. Unused video/audio data inputs grounded; TMDS outputs not connected.

PD low at power-up selects 7-bit 0x39. Wait >=200 ms before register access.

12 MHz clock drives CEC; ARC-only operation / EDID access without video clock needs bench proof.

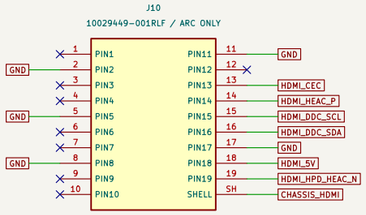
Public guides establish ARC function, not guaranteed DD+ 192 kHz carrier or Atmos passthrough.

Sheet: /arc-receiver/
File: arc-receiver.kicad_sch

Title: Open Cinema — arc-receiver

Size: A3	Date: 2026-11-04	**Rev: A11 DRAFT**
KiCad E.D.A. 10.0.5		Id: 6/9

A11 ARC STUDY — HDMI coupling, CEC, DDC and protection









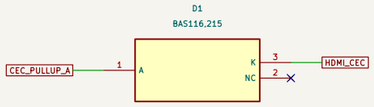








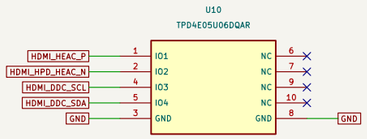


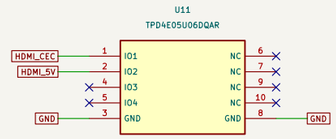












HEAC bias follows HW guide Fig17; eval variants differ. Verify bias, ESD loading and HPD on bench.

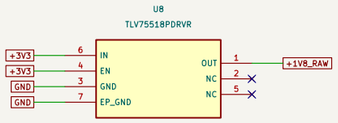
Connector-side ESD is provisioned; system ESD/CEC leakage, shell bond and back-powering not qualified.

Sheet: /arc-connector/
File: arc-connector.kicad_sch

Title: Open Cinema — arc-connector

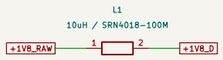
Size: A3	Date: 2026-11-04	**Rev: A11 DRAFT**
KiCad E.D.A. 10.0.5		Id: 7/9

A11 ARC STUDY — 1.8V supply filters and HDMI +5V

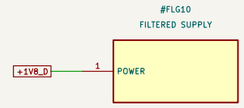


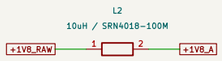




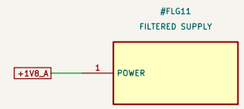






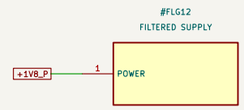


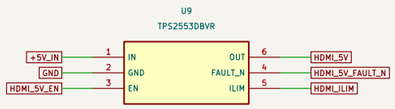












U9 ILIM=226k: approximately 101..142mA limit incl resistor tolerance (TI table). Not 75mA strap.

Sheet: /arc-power/
File: arc-power.kicad_sch

Title: Open Cinema — arc-power

Size: A3	Date: 2026-11-04	**Rev: A11 DRAFT**
KiCad E.D.A. 10.0.5		Id: 8/9

A11 ARC STUDY — local decoupling and HDMI power control







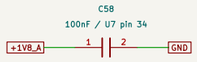








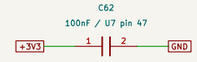




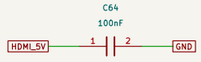


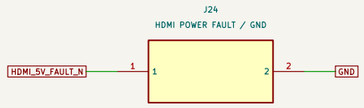


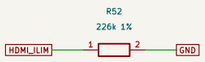
















Placement is a space study. Route decouplers at their named U7 pins with short ground vias.

Sheet: /arc-decoupling/
File: arc-decoupling.kicad_sch

Title: Open Cinema — arc-decoupling

Size: A3	Date: 2026-11-04	**Rev: A11 DRAFT**
KiCad E.D.A. 10.0.5		Id: 9/9